(iii) Let $I = \int \text{cosec}\, x dx = \dfrac{\int \text{cosec}\, x(\text{cosec}\, x - \cot x)}{\text{cosec}\, x - \cot x} dx$

$$= \int \frac{\text{cosec}^2 x - \text{cosec} x \cot x}{\text{cosec} x - \cot x} dx$$

Putting $\text{cosec}\, x - \cot x = u$, then $(\text{cosec}^2 x - \text{cosec}\, x \cot x)\, dx = du$

Thus, $I = \int \frac{1}{u} du = \log|u| + c = \log|\text{cosec} x - \cot x| + c.$

(iv) Let $I = \int \sec x dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$

Putting $\sec x + \tan x = u$, then $(\sec^2 x + \sec x \tan x)\, dx = du$

Thus, $I = \int \frac{1}{u} du = \log|u| + c = \log|\sec x + \tan x| + c$

Therefore, $\int \sec x dx = \log|\sec x + \tan x| + c.$

Thus the following are the important standard results.

(1)	$\int \tan x\, dx = \log\|\sec x\| + c$
(2)	$\int \cot x\, dx = \log\|\sin x\| + c$
(3)	$\int \text{cosec} x\, dx = \log\|\text{cosec}\, x - \cot x\| + c$
(4)	$\int \sec x\, dx = \log\|\sec x + \tan x\| + c$

Example 11.32

Integrate the following with respect to x.

(i) $\int \frac{2x+4}{x^2+4x+6} dx$ (ii) $\int \frac{e^x}{e^x - 1} dx$ (iii) $\int \frac{1}{x \log x} dx$

(iv) $\int \frac{\sin x + \cos x}{\sin x - \cos x} dx$ (v) $\int \frac{\cos 2x}{(\sin x + \cos x)^2} dx$

Solution

(i) Let $I = \int \frac{2x+4}{x^2+4x+6} dx$

Putting $x^2 + 4x + 6 = u$, then $(2x+4)\, dx = du$

Thus, $I = \int \frac{du}{u} = \log|u| + c = \log|x^2 + 4x + 6| + c$

Therefore, $\int \frac{2x+4}{x^2+4x+6} dx = \log|x^2 + 4x + 6| + c.$

11_11th-Maths-Vol2-Unit11.indd 205 16-12-2022 11:07:29